Beyond Detection of Explainable AI for Pneumonia Diagnosis

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***Abstract*—Pneumonia is a serious respiratory disease that can be detected through chest X-ray imaging. This study applies deep learning techniques to classify chest X-ray images and identify pneumonia cases automatically. The project aims to develop a high-accuracy system for detecting thoracic diseases that also uses explainable AI (XAI) to highlight abnormal regions, improving transparency and clinical trust in automated diagnosis.**

***Index Terms*—Pneumonia Detection, Chest X-ray, Deep Learning, Explainable AI, Medical Imaging**

I. Problem Statement and Importance

Pneumonia detection from chest X-ray images using deep learning faces several challenges:

- **Black-box predictions:** Many deep learning models achieve high accuracy, but their decisions are difficult for clinicians to interpret, limiting trust in clinical use.
- **Limited localization explainability:** Explainability methods such as Grad-CAM and SHAP can highlight important regions, but their reliability varies across models and datasets.
- **Dependence on costly annotations:** Accurate pneumonia localization often requires bounding boxes or pixel-level labels, which are expensive to obtain. Weakly supervised learning reduces this burden, but localization remains challenging.
- **Generalization and data quality issues:** Chest X-ray datasets may contain class imbalance, noisy labels, and patient-split leakage, reducing robustness and real-world performance.

II. Algorithms and Models

The study uses 12 models for pneumonia detection and interpretability, grouping them by their feature extraction method and computational efficiency.

A. Baseline and Traditional Machine Learning

To establish a performance floor, we implement **K-Nearest Neighbors (KNN)**, a non-parametric algorithm used to evaluate the clustering of pneumonia features in a high-dimensional feature space. Additionally, a custom **Convolutional Neural Network (CNN)** is developed as a baseline for basic spatial hierarchies without the influence of pre-trained weights.

TABLE I
Summary of Final Evaluated Models

Category	Specific Models
Baselines	KNN, Custom CNN
Residual	ResNet18, ResNet50, ResNet152 v2
Dense	DenseNet-121
Mobile/Efficient	MobileNetV2, MobileNetV3, EfficientNet-B0
Advanced	ViT-B16
Framework	XAI-ICP

B. Residual and Dense Deep Networks

We utilize the ResNet family, including **ResNet18**, **ResNet50**, and **ResNet152 v2**, to analyze how increasing depth and skip connections impact the identification of subtle pathological features. Furthermore, **DenseNet-121** is employed to maximize feature reuse through dense connectivity, which is highly effective for medical imaging where gradient flow is critical.

C. Efficient and Mobile Architectures

For computational efficiency, we implement **MobileNetV2** and **MobileNetV3**, which utilize depthwise separable convolutions to reduce parameter count. We also include **EfficientNet-B0**, which applies compound scaling to balance resolution, depth, and width for optimal classification performance.

D. Vision Transformers and Explainability

To capture long-range global dependencies that standard convolutions might miss, we implement the **Vision Transformer (ViT-B16)**. Finally, the **XAI-ICP** (Explainable AI - Integrated Class Probability) framework is integrated across all architectures. This framework provides the necessary interpretability through heatmaps and attribution scores to ensure the models are making decisions based on relevant clinical indicators.

III. Data Specifications

A. Dataset Overview

- **Name:** ChestX-ray14 (NIH Chest X-ray Dataset)
- **Source:** Kaggle (Community-hosted version)
- **Access URL:** https://www.kaggle.com/datasets/nih-chest-xrays/data

B. Dataset Details

- **Total Images:** 112,120 frontal-view chest X-rays
- **Unique Patients:** 30,805
- **Image Format:** PNG
- **Resolution:** 1024×1024 pixels
- **Label Type:** Multi-label classification
- **Disease Categories:** 14 thoracic conditions

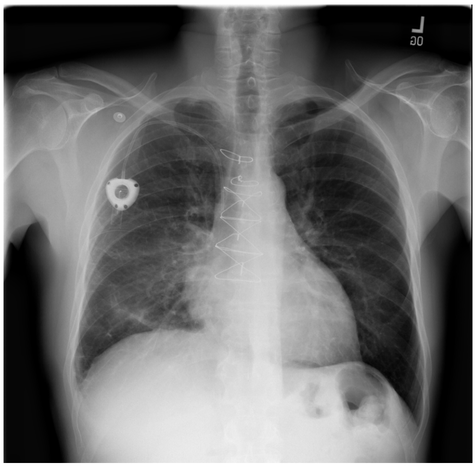

Fig. 1. Example chest X-ray image from the ChestX-ray14 dataset showing lung structure.

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